



How do large-language models respond to moral dilemmas? Insights from the defining issues test

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Abstract

This study examined how various large language models (LLMs) addressed moral reasoning dilemmas using the Defining Issues Test (DIT-2). It compares responses across nine major LLM platforms (ChatGPT, Grok, Claude, Gemini, Mistral, Microsoft Copilot, Perplexity, Llama, and DeepSeek AI). The study also compared LLM and human responses and examined how various prompt structures and interventions, such as introducing influential moral figures, modified LLM responses. The findings showed significant differences in moral reasoning capacities among the 32 LLM models examined, with models varying in their ability to generate responses that follow established moral reasoning patterns. In post-conventional reasoning, high-performing models outperformed human norms, with mean scores ranging from 34 to 82 across models. LLMs demonstrated significantly higher post-conventional reasoning than humans ($M=64.25$ vs. $M=34.86$) and N2 scores ($M=63.50$ vs. $M=34.26$). Following this pattern of LLM superiority, personal interest reasoning was lower in LLMs ($M=11.12$ vs. $M=25.77$), as was maintaining norms reasoning ($M=23.25$ vs. $M=33.79$). While differences between LLM groups were not statistically significant, effect sizes indicated distinct moral reasoning profiles. In addition, prompt modifications had a selective effect, significantly lowering personal interest reasoning while slightly increasing maintaining norms reasoning, and lower P scores while holding N2 scores constant. Finally, interventions involving historical moral figures were effective in shifting responses from post-conventional to more maintaining norms reasoning. This study improves understanding of LLMs' moral reasoning abilities and provides valuable insights for value alignment, responsible deployment, AI safety, and the design of ethical AI systems.

Keywords Large language models (LLMs) · Moral reasoning · Defining issues test (DIT-2) · Artificial Intelligence ethics · Neo-kohlbergian theory

1 Introduction

A major advancement in AI in recent years has been the evolution of large language models (LLMs) [10]. LLMs, such as OpenAI's ChatGPT, Grok, Claude, Gemini, Mistral,

Microsoft Copilot, Perplexity, and Llama, can produce very accurate and contextually relevant results, perform complex tasks such as problem-solving, content creation, customer support, language translation, code generation, personal assistant, sentiment analysis, and even multimodal

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integration, making them adaptable tools for a wide range of applications [1, 3, 5, 24, 32, 38, 57, 59]. LLMs are trained in several stages, including pre-training, supervised fine-tuning, and Reinforcement Learning from Human Feedback (RLHF) [31, 33, 36, 58]. As a result, through these various stages, the LLM model develops predictive capabilities in syntax, semantics, and human language structures by learning statistical patterns across billions of parameters [36].

While the benefits of LLMs in natural language processing and automation are well known, their ability to contribute to complex fields such as morality and moral reasoning could open up new avenues for understanding AI ethics and improving decision-making ([18]; 2025; [41, 50, 51, 54]).

Moral reasoning has been a topic of interest among philosophers, educators, sociologists, and psychologists for many years [47]. Moral reasoning is the cognitive and emotional process that happens when individuals deliberate about what is morally right or wrong in a particular situation [28, 47]. Moral reasoning can involve principles, rules, and values as guides for ethical decision-making [27]. For example, a newspaper reporter must decide whether to publish a damaging story about a political candidate that could affect an election outcome [17]. The most well-known understanding of moral reasoning is rooted in Lawrence Kohlberg's classic theory [28], now advanced by neo-Kohlbergian enhancements, to emphasize moral schemas instead of stages of development. From this perspective, global or bedrock moral reasoning at its best is 'post-conventional' for being capable of sophisticated and nuanced reasoning that is not restrained by faulty communal norms or by too much personal interest. More contextual levels of moral reasoning are explained by the same theory as relating to intermediate concepts and specific moral codes [43]. Neo-Kohlbergian moral reasoning is described further in later sections as being central to the article.

Moral reasoning is important for both humans and AI systems for different reasons. First, moral reasoning is essential for ethical decision-making, social cooperation, and moral development throughout the lifespan [27]. On the other hand, moral reasoning capabilities are critical for value alignment, responsible deployment, and AI safety in AI systems, particularly LLMs, which increasingly interact with moral content and decision-making contexts. As LLMs are integrated into healthcare, legal, educational, and policy contexts, understanding how these systems process ethical information becomes essential for ensuring they augment rather than undermine human moral judgment [18, 25].

The evaluation of LLM moral reasoning is crucial for AI ethics, safety, and value alignment. First, it assesses the alignment of AI systems with human moral values, a key challenge in AI safety research [50, 51]. For example, models with weak or inconsistent moral reasoning cause major

risks when used in sensitive situations, which might amplify harmful biases, provide unethical guidance, or undermine public trust in AI technologies. Second, understanding how LLMs process moral information helps to make responsible deployment decisions, such as which applications are suitable for AI assistance, how much human oversight is required, and where AI moral reasoning capabilities are insufficient. Third, evaluating LLM moral reasoning can help identify potential misalignment between training objectives (predicting human text) and normative ethical standards, which reveals gaps between statistical pattern learning and principled ethical judgment [54]. Finally, systematic evaluation of AI systems using credible theories (neo-Kohlbergian moral reasoning) and validated instruments (Defining Issues Test) has the potential to benchmark progress toward ethically capable AI systems.

Despite growing interest in AI ethics, there are still major gaps in understanding LLM moral reasoning. First, most previous research only looks at one or a few LLM models, which limits systematic comparisons across architectures, training approaches, and platforms. This narrow scope makes it difficult to identify architecture-specific patterns or reach general conclusions. Second, many studies use random ethical scenarios or hypothetical dilemmas that lack validated assessment instruments, making it difficult to interpret and compare results across studies. Third, little is known about the adaptability of LLM moral reasoning, specifically whether and how prompt engineering or contextual interventions can systematically shift moral reasoning patterns. As a result, these gaps are critical for understanding LLMs' moral reasoning alignment and both the risks (vulnerability to manipulation) and opportunities (potential for alignment through guidance) of using LLMs in morally relevant contexts.

In response to the gaps identified above, the study reported in this article explored LLMs' moral reasoning capacities, examining their current strengths and weaknesses in order to improve the understanding of how AI systems process ethical information. It also compared LLM-generated responses to human responses to identify significant differences, which provides insights that can support educators and developers in developing AI systems that support rather than replace human moral reasoning. This study additionally examined the effects of prompt engineering on LLM moral reasoning responses. This approach allowed the current study to examine how different prompt structures influence LLM's moral reasoning responses and identify differences in LLM's moral reasoning patterns.

Moreover, interventions were conducted to examine the possibility that LLMs could modify their responses based on new information or guidance. The interventions involved providing LLMs with information about moral exemplars

and examining whether exposure to them would change the models' responses to moral dilemmas (situations in which an individual must choose between two or more opposing options). These interventions provide valuable information about the flexibility and adaptability of LLMs in moral reasoning across different contexts.

The following research questions are the focus of the current study:

1. How do different LLM models compare in their capabilities to generate responses consistent with moral reasoning patterns?
2. To what extent do LLM moral reasoning responses align with human responses?
3. How do different prompting strategies affect LLMs' moral reasoning responses?
4. How do interventions, including the introduction of moral exemplars or new information, modify LLM moral reasoning patterns?

Understanding LLM's moral reasoning is important for various stakeholder groups. For example, for AI developers, systematic evaluation informs design decisions, training procedures, and safety measures. For policymakers, it provides evidence for regulatory frameworks and deployment guidelines. In addition, for educators, it clarifies which AI systems can appropriately support moral education and which may introduce problematic biases. Moreover, for researchers, it establishes validated methods for assessing AI moral capabilities and tracking progress toward value-aligned systems. As a result, this study contributes to these goals by providing a comprehensive evaluation across multiple models that may improve alignment between AI and human moral judgment. In summary, the study's findings could potentially help developers, policymakers, and researchers in ensuring that LLM-generated moral reasoning evolves in a transparent, contextually aware, and consistent manner with human and societal values.

2 Theoretical framework

Neo-Kohlbergian moral reasoning, as part of a broader four-component model (FCM) of moral functioning, underpins the study. As mentioned earlier, this allows for systematic comparisons of human and AI moral reasoning patterns using empirically validated assessment tools (Defining Issues Test).

Kohlberg's original work was criticized for emphasizing unsupportable hard moral stages and for emphasizing moral reasoning when there is evidence to suggest this often did not translate into moral action. James Rest developed

a neo-Kohlbergian position known as the Four-Component Model (FCM), which expands morality beyond purely moral reasoning [35, 45]. The FCM involves four psychological components working together (Fig. 1).

First, moral sensitivity is the awareness that our actions affect others. This involves empathy, role-taking, and recognizing ethical issues. This relates to moral reasoning by identifying situations requiring moral evaluation. Individuals who lack moral sensitivity may fail to recognize moral issues ([35]; [45]).

Second, moral judgment is the process of distinguishing between right and wrong. Obviously, moral reasoning features here and works on three levels: codes of conduct, intermediate concepts, and bedrock schemas. Codes of conduct are specific rules that require minimal interpretation; intermediate concepts apply values such as honesty to specific situations and are evaluated by the Intermediate Concepts Measures (ICM); and bedrock schemas are general principles assessed by the Defining Issues Test (DIT) ([35]; [45]).

Third, moral motivation prioritizes moral values over personal gain or social approval and supports the execution of moral decisions. It involves the motivation to act morally. One penalty of emphasizing moral reasoning in Kohlberg's theory was limited acknowledgement of this crucial component for moral agency ([35]; [45]).

Finally, moral character is the ability to act on moral values and judgments, which includes ego strength, perseverance, integrity, resilience, strength of conviction, and courage. Moral character is the execution phase, where reasoning and motivation translate into action. This component acknowledges that moral behavior requires not just knowledge and motivation, but also the personal strength to act despite obstacles or opposition. The FCM shows that moral reasoning is not isolated but is related to moral sensitivity, judgment, motivation, and character components. A deficiency in any one component can lead to moral failure, even if the other components are strong ([35]; [45]).

Neo-Kohlbergian theory recognizes the relevance of personal and cultural influences on moral reasoning [44, 46]. Central to neo-Kohlbergian ideas of moral reasoning are moral schemas (replacing Kohlberg's original stages). Moral schemas are "general knowledge structures residing in long-term memory" that structure moral reasoning by providing frameworks for interpreting moral situations and determining appropriate responses. As such, moral schemas are general frameworks for interpreting moral situations. These schemas vary in complexity and abstraction. For example, less sophisticated schemas rely on concrete rules and immediate consequences, whereas more advanced schemas incorporate abstract principles and universal ethical considerations. At the most general and non-contextual

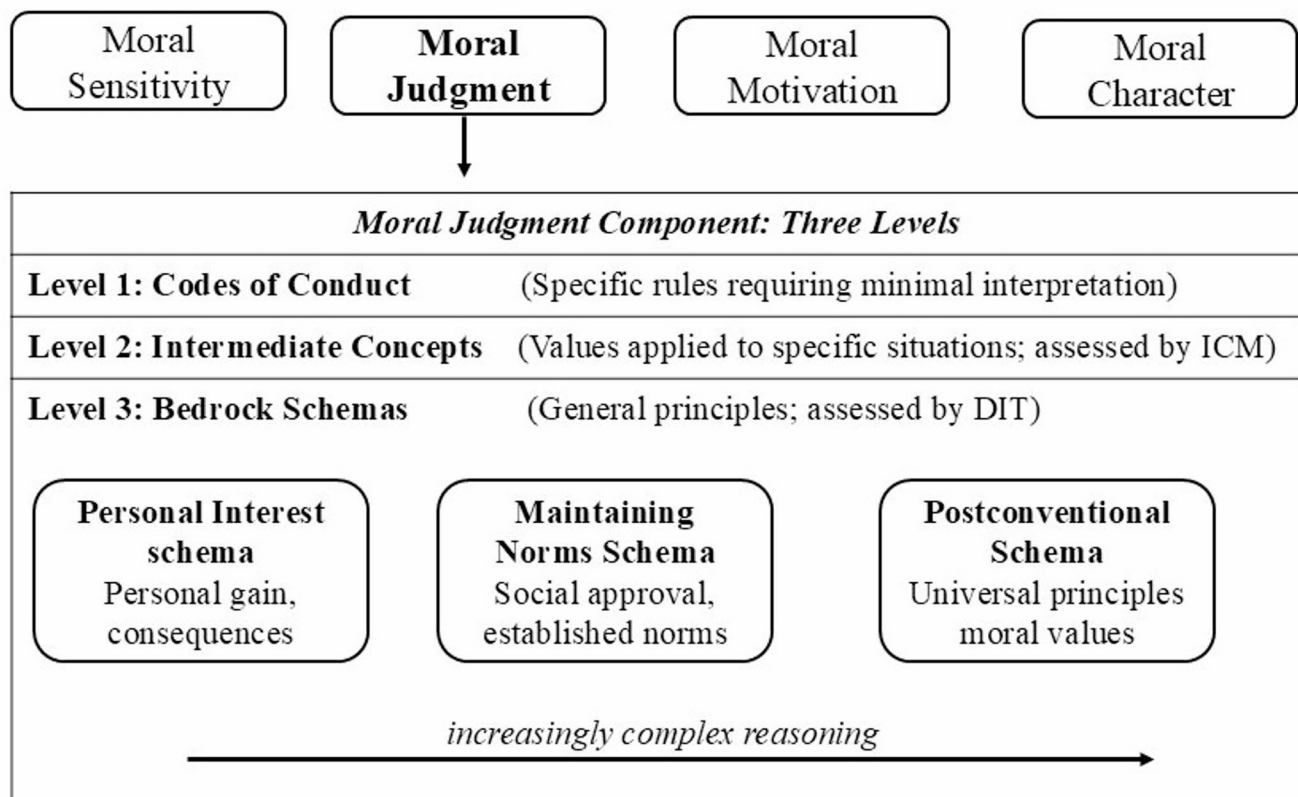


Fig. 1 The four-component model

level are so-called bedrock moral schemas, also consisting of three levels of reasoning from worse to better: personal interest (individual gain and immediate consequences), maintaining norms (social order, laws, and conventions), and post-conventional (universal ethical principles like justice and human rights). Post-conventional reasoning is the most advanced form of moral reasoning in the neo-Kohlbergian framework. It is defined by principled reasoning founded on universal ethical principles such as justice, human rights, and human dignity, rather than conventional norms or personal interests. Furthermore, each schema represents a unique method of conceptualizing moral issues and determining appropriate responses [42]. Each schema represents a unique method of conceptualizing moral issues and determining appropriate responses [42]. Measurement of bedrock moral reasoning is via the Defining Issues Test (DIT) that is described in later sections. The measure activates moral schemas based on the three levels [42, 46].

Moral reasoning is especially important in the context of AI, particularly LLMs, because it provides the cognitive foundation for moral judgment. While moral sensitivity involves recognizing ethical issues and moral motivation involves emphasizing values, moral reasoning requires a capacity to apply abstract ethical principles to novel, complex situations. This is one of the challenges that AI systems

face when faced with complex scenarios without human direct guidance [2, 20].

Emphasizing moral reasoning is theoretically valid and methodologically useful. In theory, AI systems such as LLMs can perform sophisticated pattern recognition and logical analysis without the emotional engagement or experiential learning that characterizes human moral development, which makes moral reasoning the most accessible component for systematic study [50, 51]. Methodologically, the extensive research foundation in moral reasoning, including the comprehensive DIT database at the Center for the Study of Ethical Development, provides strong comparative benchmarks for assessing AI-human moral alignment [19, 56].

This focus becomes increasingly important as AI systems are deployed in situations that need moral judgment [13, 53]. Moreover, understanding how LLMs process moral reasoning makes it possible to better assess their critical ethical challenges and limitations, and make sure these systems support, not violate, human moral agency [26]. Eventually, while all four FCM components are required for full moral functioning [45], moral reasoning is the critical starting point for understanding how AI systems can be created to enhance human moral judgment processes.

The neo-Kohlbergian framework has strong connections to general ethical theories. Post-conventional reasoning can include both consequentialist and deontological considerations, recognizing that principled moral judgment can involve weighing consequences (utilitarian reasoning) while maintaining fundamental rights and duties. In neo-Kohlbergian theory, the highest levels of moral reasoning require the integration of multiple ethical considerations rather than an exclusive commitment to a single normative framework [43]. This theoretical flexibility makes the DIT-2 suitable for evaluating AI systems that may need to reason across multiple ethical frameworks depending on the context. In addition, the neo-Kohlbergian framework's emphasis on contextual judgment is consistent with recent research on *phronesis* or practical wisdom (contextual integrative thinking that enables optimal decisions in a variety of situations) [29]. Moreover, moral reasoning is embedded in larger networks of psychological indicators, with empathy and purpose operating as foundational predictors [21]. This theoretical perspective informs our approach to assessing LLM moral reasoning through examination of reasoning sophistication (post-conventional schema), adaptability to contextual interventions, and consistency across different moral dilemmas.

2.1 Literature review

This section explores previous research on LLM moral reasoning abilities, human-AI comparisons, and the effects of various prompting strategies on moral reasoning. In the context of AI, bias is defined not only as a statistical deviation but also as a systematic skew in model outputs that can lead to unfair, prejudicial, or ethically inconsistent treatment of specific groups or concepts [12]. This study defines bias as systematic deviations from human normative patterns in moral reasoning. We distinguish three types of bias: (1) statistical bias, which over-represents specific moral reasoning patterns; (2) normative bias, which is a systematic preference for particular values or principles; and (3) contextual bias, which is differential reasoning based on in-context learning and interventions. Understanding these biases is critical for determining LLMs' suitability for use in contexts that require moral judgment.

3 LLM moral reasoning and adaptability through training

Research demonstrates that LLMs trained with RLHF can follow sophisticated normative concepts and avoid undesirable outcomes when given clear ethical instructions [14]. Larger models show improved capacity to reduce bias

across multiple benchmarks, and unlike earlier models that required algorithmic adjustments, modern LLMs can be guided through natural language instructions [4, 15, 16]. This adaptability increases with model size and the level of RLHF training [14]. In addition, interventions using external moral content demonstrate that LLMs possess in-context learning and chain-of-thought reasoning capabilities, which enable LLMs to modify responses based on new information. For example, when ChatGPT was provided with Martin Luther King Jr.'s "Letter from Birmingham Jail," its responses shifted toward more post-conventional reasoning [20]. On the other hand, it is important to note that research also found that historic figures were less effective than peer exemplars in promoting moral development among human learners [23].

4 Model architecture and training effects

Research indicates that both architectural and training factors significantly influence LLMs' moral reasoning abilities. Larger models consistently demonstrate more sophisticated moral reasoning patterns, with multi-billion parameter models outperforming smaller variants in terms of contextual ethical judgment [14, 34]. However, size alone is insufficient; training methodology is equally important.

Pre-training with diverse, high-quality data increases moral knowledge and improves understanding of ethical principles [4]. Models trained primarily on small datasets show limited moral reasoning capacities and are more vulnerable to problematic biases. Moreover, supervised fine-tuning enhances models' ability to follow ethical guidelines and respond appropriately to moral questions. Most significantly, RLHF appears to be critical for aligning model outputs with human moral judgment [15]. In addition, models trained with extensive RLHF demonstrate an improved ability to recognize ethical considerations, avoid harmful outputs, and respond to moral guidance via natural language instructions [30]. However, RLHF introduces biases stemming from the specific values and perspectives of human raters, raising concerns about whose moral standards are being encoded [11]. These findings show that LLM moral reasoning is not an emergent property of scale, but rather the result of deliberate design and training choices, which suggests that targeted interventions can improve moral reasoning.

5 Theoretical approaches to evaluating LLM moral reasoning

Previous research has used a variety of theoretical frameworks to examine LLMs' moral reasoning, highlighting various aspects of moral judgment. Moral Foundations Theory (MFT) evaluates LLM performance across five moral foundations, typically revealing a preference for care and fairness while overlooking loyalty, authority, and sanctity [50]. Utilitarian and deontological frameworks highlight variations in reasoning behavior in trolley problem scenarios, revealing inconsistent application of consequentialism and duty-based reasoning [40]. In addition, virtue ethics examines LLMs' understanding of character traits but finds them lacking in context-specific applications [52]. As mentioned earlier, many studies use unstandardized hypothetical dilemmas, which complicate comparisons. However, the neo-Kohlbergian approach taken in the present study differs by emphasizing sophisticated moral reasoning and by having validated assessments used in multiple and varied contexts over many decades.

6 Human-AI alignment and comparative performance

Studies comparing LLM to human moral reasoning show both similarities and significant differences. While larger LLMs and those with RLHF training generally align more closely with human judgment, they differ significantly in how they weigh specific factors like norm violations and unavoidable harm [37]. For example, GPT-4 has a stronger tendency to consider actions morally acceptable when harm is unavoidable, whereas humans show more sophisticated contextual assessment. In addition, some LLMs are more likely than humans to see abnormal events as causal [37].

Empirical assessments using standardized moral reasoning instruments reveal significant variation among models. Han, [20] discovered that ChatGPT (May 2023) scored 45.83 in post-conventional reasoning, which was slightly lower than the human undergraduate median (50.00) and ranked in the 40th percentile. In contrast, comparative studies using the Defining Issues Test revealed that GPT-4 achieved post-conventional scores comparable to graduate students (P-score 55.68), whereas GPT-3 performed near random baseline levels [55]. These variations indicate that model architecture, training data, and fine-tuning approaches have a significant impact on moral reasoning abilities [55]. Importantly, LLMs can approximate human moral judgments in many cases, but they may prioritize different factors or interpret moral scenarios differently, which

emphasizes the need for careful evaluation when deploying these systems in contexts requiring moral reasoning [37].

Identified gaps and opportunities in the literature informed the study's research design. While previous research has shown that LLMs have varying moral reasoning capacities depending on model size, training methods such as RLHF, and prompt engineering [48], most studies have focused on extremely limited model numbers or used less comprehensive assessment tools. Informed by neo-Kohlbergian theory, this study examines 32 LLM models using DIT-2, enabling the identification of architecture- and platform-specific patterns. The study also compares LLMs to large human datasets, systematically testing prompt engineering effects and implementing intervention strategies based on historical moral exemplars. Finally, unlike most previous studies that used the bDIT [20] and DIT-1 [55], the current study used the DIT 2 - the most recent and widely used version. DIT 2 includes the N2 score and additional indices that were not available in previous versions.

6.1 Methodology

This study used comparative analysis and an experimental design to evaluate LLM's moral reasoning abilities.

7 Participants

7.1 Large language models (LLMs)

The study examined 32 different LLM model variants across nine major platforms: ChatGPT, Grok, Claude, Gemini, Mistral, Microsoft Copilot, Perplexity, Llama, and DeepSeek AI.

7.2 Human data

The human data were derived from the Defining Issues Test-2 (DIT-2) normative dataset, which includes responses from 73,740 participants and was published as a norms article by the Center for the Study of Ethical Development [19]. The sample had a mean age of 23.11 years ($SD = 7.87$, age range 12–95 years), which indicates a relatively young population with moderate age variability. Regarding political views, the distribution showed that 31.3% identified as liberal (7.4% very liberal + 23.9% somewhat liberal), 27.8% as neither liberal nor conservative, and 40.9% as conservative (30.8% somewhat conservative + 10.1% very conservative), which shows a slight conservative lean in the sample. The gender distribution was nearly equal, with 49.7% male ($n = 36,678$) and 50.3% female ($n = 37,062$) participants. For educational levels, the majority were undergraduates

(67.4%), followed by graduate or professional degree holders (17.1%), students with Grade 1–12 education (10.5%), and vocational/technical or junior college students (5.0%). In addition, all participants (100%) were U.S. citizens and identified English as their primary language, which shows a homogeneous sample in terms of citizenship status and primary language. The dataset includes data from 2011 to 2020, with 53.5% completing the paper-based version and 46.5% completing the online version of the test [19].

7.3 Measures

7.3.1 Defining issues test (DIT)

The Defining Issues Test-2 (DIT-2) is a widely validated instrument for assessing moral reasoning that consists of five moral dilemmas. The full version of DIT-2 was used. This is computed and scored using five dilemmas. For each dilemma, respondents select what the main character should

do (action choice), rate the importance of 12 items, and rank the four most important items. An example from the Famine story is: “Is Mustaq Singh courageous enough to risk getting caught for stealing?” This item could be rated on a scale from “Great” to “No,” depending on how important the respondent considers courage in this context [8, 56]. This structure enables DIT-2 to evaluate both action preferences and moral reasoning patterns [56].

DIT-2 generates four primary indices that represent schema usage likelihood. The DIT-2 schema scores represent the extent to which various types of moral reasoning are used and are categorized into three schema scores: Personal Interests, Maintaining Norms, and Postconventional. These schema scores reflect how individuals prioritize personal gain, social rules, or broader moral ideals (universal principles) in moral reasoning. The personal interest schema (0–100) indicates the lowest level of moral reasoning (self-interest, individual benefits, and social approval). The maintaining norms schema score (0–98) corresponds to conventional rule-based reasoning. The post-conventional schema score (0–95) indicates sophisticated and principled reasoning. The N2 score (0–95) is a more recent indicator that assesses both post-conventional prioritization and personal interest de-emphasis, which offers a more nuanced measure of moral development than the traditional P-score. In addition to primary schema indices, DIT-2 includes additional measures to characterize moral reasoning patterns and ensure data validity (see Table 1). These measures ensure that responses reflect valid moral reasoning rather than being random or fabricated [8]. It is important to note that DIT-2 is a copyrighted measure, and scores were calculated (the scoring process is confidential) by the University of Alabama’s Center for the Study of Ethical Development.

7.4 Procedure

The study was performed in several phases. First, the five dilemmas from the DIT-2 measure were provided to LLM platforms to evaluate their capabilities to generate responses consistent with moral reasoning patterns. Second, responses across different LLMs were compared to evaluate their differences. Third, LLMs and human responses were compared to identify similarities and differences in moral reasoning. Fourth, LLM responses were examined to determine how they differed when using a standard prompt from the measure versus an additional instruction, such as “Please ensure your answer is unbiased.” The unbiased prompt was included because previous research in prompt engineering shows that explicit instructions to avoid bias can reduce certain systematic response tendencies, particularly social desirability effects [14, 20]. In AI alignment research, general “unbiased” instructions usually reveal a model’s default

Table 1 DIT-2 indices summary

	Index	Range	Definition
Primary scores	Personal interests	0–100	Self-interested reasoning
	Maintaining norms	0–98	Conventional reasoning
	Post-conventional (PSCORE)	0–95	Principled reasoning
	N2 score	0–95	Post-conventional emphasis + PI de-emphasis
Type & Consistency	Type indicator (TYPENEW)	1–7	Dominant schema type
	Consolidation (CONSTRAN)	1–2	Schema stability (1 = transitional, 2 = consolidated)
Supplementary indices	Utilizer (U)	–1 to 1	Action-reasoning consistency
	Humanitarian liberalism (HUMLIB)	0–5	Humanitarian values alignment
	Religious orthodoxy (CANCER10)	1–9	Religious content (↓score=↓religious)
	Cannot decide (NUMCD)	1–5	Indecision frequency
	Anti-social score (ASCORE)	0–16	Anti-establishment attitudes
Validity checks	Meaningless items (MSCORE)	0–16	Invalid if > 10
	Consistency check (TOTCC)	> 200 = purged	Multi-indicator reliability

training biases or cause it to revert to a “safe,” conventional middle ground [6]. Finally, the study assessed whether LLMs could learn from interventions (see below). Figures were chosen to represent different ethical themes to test context sensitivity. As an example, in story 1, Jean Valjean represents the conflict between legal duty (norms) and human survival (post-conventional justice). Guidance from historical or influential figures was provided to see if LLMs modified their answers in response to new information.

7.5 Interventions

The adoption of moral figures into interventions is based on LLMs’ emerging abilities of in-context learning and chain-of-thought reasoning, which allow them to learn from exemplars and update their reasoning processes [20]. Moral exemplar stories provide concrete examples of sophisticated reasoning that serve as effective learning inputs, similar to how “real-life” models outperform hypothetical dilemmas in promoting moral development [22, 23]. The current study incorporated historical moral figures into DIT-2 dilemmas to investigate how exposure to such moral exemplars influences LLM moral reasoning patterns, without presuming the direction of potential changes. Each intervention included two types of prompts. Prompt A provided a clear introduction to the influential figure’s principles and a guiding question to respond to the dilemma. Prompt B replaced the original character with influential figures, which allows LLMs to reason directly by adopting a persona within the story’s context. This methodology enabled a systematic investigation into whether and how moral role models influence LLM moral reasoning patterns.

1. *Famine story: Famine (Stealing food for survival).*

Intervention

Jean Valjean (from Victor Hugo’s *Les Misérables*) was introduced using both prompt types to address the father as the decision-maker in the famine situation. Jean Valjean, who famously stole bread to save his family from starvation, represented the moral complexities of surviving under unjust systems.

2. *Reporter Story: (Revealing a candidate’s past misdeeds)*

Intervention

Ida B. Wells, a historical figure known for her dedication to journalistic ethics and social justice, was introduced through the dual-prompt approach to influence the reporter’s decision-making process. Ida B. Wells, an investigative journalist and activist, approached the dilemma from

the perspective of holding public figures accountable while considering the greater good of society.

3. *School Board Story: (Deciding whether to hold an open meeting amidst threats)*

Intervention

Ruth Bader Ginsburg, an advocate for justice and public advocacy, was introduced to influence Mr. Grant’s decision-making using both prompt methodologies. Justice Ruth Bader Ginsburg’s legacy of balancing equality, fairness, and legal precedent provided a different perspective.

4. *The Cancer Dilemma Story: (Mrs. Bennett’s request for more medication):*

Intervention

Pope John Paul II, known for his commitment to the sanctity of life, was introduced through the dual-prompt framework to influence the decision on whether to administer an overdose to a patient in pain.

5. *Demonstration story: (student protests against the US military actions)*

Intervention

Both prompt approaches introduced Martin Luther King Jr., an advocate for peaceful protest and nonviolent resistance, as the decision-maker.

Each figure was chosen based on how well their historical actions, ethical principles, and documented reasoning aligned with the core moral dilemma in each DIT-2 story (designed based on prior research on historical moral exemplars). For example, Jean Valjean’s experience stealing food for survival aligned with the famine dilemma; Ida B. Wells’ investigative journalism matched the reporter’s ethical conflict; Ruth Bader Ginsburg’s judicial experience with balancing advocacy and constraints fit the school board scenario; Pope John Paul II’s end-of-life teachings corresponded with the cancer story (religious moral figure related to item 10 on God’s role in ending life), and Martin Luther King Jr.’s civil disobedience philosophy addressed the demonstration context. This matching approach ensured that each exemplar could provide contextually relevant moral reasoning.

Each intervention was carefully performed across all LLM models. The intervention process consisted of providing background information about the historical figure, explaining their moral philosophy or ethical stance, and re-presenting the dilemma with the historical figure as the decision-maker. The baseline responses were for pre-intervention and compared to post-intervention results. This method enabled researchers to determine if LLMs could

incorporate new moral perspectives and modify their moral reasoning patterns.

7.6 Data analysis

Data analysis was conducted using SPSS version 29. Descriptive statistics were used to summarize the various indices scores for LLMs in baseline, unbiased and intervention phases. To conduct a comparative analysis across LLM models, ANOVA was used to compare responses to identify differences in moral reasoning abilities.

To compare LLM responses with human data, Mann-Whitney U and t-tests were used to compare the LLM responses to the data from human participants. T-tests were also used to compare the impact of various prompts on LLM responses. In the experimental intervention analysis, pre- and post-intervention scores were compared via ANOVA to determine whether the LLMs showed in-context learning or

adaptability. Furthermore, given the large sample size and multiple comparisons in the study, effect size calculations were performed to provide additional information.

8 Results

8.1 Descriptive statistics results for initial LLM responses

The schema distributions and N2 scores for LLM responses are shown in Table 2, which contains a total of 32 LLMs.

The descriptive statistics for the initial responses across all 32 LLM models revealed substantial variations in moral reasoning abilities. The Personal Interest Schema (range: 0-100) had scores ranging from 0 to 24, with an average of 11.12 (SD=6.66). The majority of models (18 out of 32, representing 56%) scored below the mean, which indicates

Table 2 Schema distribution and N2 scores across LLMs for initial responses

LLMs model	Personal interest (Score range: 0-100)	Maintaining norms (Score range: 0- 98)	Post conventional (Score range: 0-95)	N2 score (Score range: 0-95)
ChatGPT- 4.5	24	10	64	58.96
ChatGPT- O3	0	22	78	75.85
ChatGPT- 4.1	4	40	54	57.71
ChatGPT-O4-Mini	20	44	36	42.48
Claude Haiku 3.5	12	16	66	52.73
Claude Opus-3	12	14	72	68.12
Claude Opus-4	6	24	70	69.38
Claude Sonnet- 3.7	8	10	82	74.98
Claude Sonnet- 4	8	20	72	71.43
Copilot Quick Response	14	10	74	63.83
Copilot Think Deeper	10	30	56	57.58
DeepSeek-Deep Think R1	8	26	66	64.75
DeepSeek-V3	18	16	66	62.67
Gemini 2.5 flash	14	32	54	60.02
Gemini 2.5 pro	6	10	82	72.86
ChatGPT-4.1-mini	24	22	50	51.88
ChatGPT-4O	24	4	68	60.27
Grok- deep Search	12	36	46	57.02
Grok-Deeper Search	20	16	64	63.88
Grok -Think	10	50	40	50.97
Grok-4	6	20	74	72.48
Llama 4	2	22	76	75.41
Mistral 10x speed	18	48	34	46.02
Mistral pure thinking	8	18	74	69.45
Mistral think	16	36	48	54.40
ChatGPT o4-mini-high	4	12	78	73.36
Perplexity Labs	4	14	82	76.42
Perplexity Research	6	44	48	53.22
Perplexity Search	6	48	46	56.60
ChatGPT 5 Thinking	6	12	82	75.44
ChatGPT-5	16	4	78	67.59
Claude Opus 4.1	10	14	76	74.15

that most LLMs showed relatively low reliance on personal interest considerations in their moral reasoning.

The Maintaining Norms Schema (range: 0–98) demonstrated greater variability, with scores ranging from 4 to 50 and a mean of 23.25 (SD=13.56). This broad distribution indicates important differences among models in their tendency to prioritize established rules and social conventions.

The Post-Conventional Schema (range: 0–95), which represents the highest level of moral reasoning, produced scores ranging from 34 to 82, with a mean of 64.25 (SD=14.53). In particular, 25 out of 32 models (78.1%) received scores ≥ 50 , which shows that the vast majority of LLMs showed sophisticated moral reasoning patterns consistent with post-conventional reasoning.

The N2 scores (range: 0–95), which provide a comprehensive measure of moral development by taking into account both post-conventional prioritization and personal interest rejection, ranged from 42.48 to 76.42, with an average of 63.50 (SD=9.56). The relatively narrow standard deviation compared to other variables (compared to human data ranging from 14.74 to 16.64) suggests that models perform more consistently on this comprehensive moral reasoning index.

These findings suggest that LLMs show moral reasoning patterns consistent with advanced developmental schemas,

which indicates a preference for principled reasoning over traditional rule-following or self-interested reasoning. In other words, the distribution indicates that most models successfully internalized sophisticated ethical frameworks during their training processes (Fig. 2).

Detailed analysis of individual model families reveals unique performance patterns and levels within each group. The results show that the ChatGPT family's post-conventional scores range from 36 (O4-Mini, lowest) to 82 (ChatGPT 5 Thinking, highest), and N2 scores range from 42.48 to 75.85. In fact, newer models, such as O3 and ChatGPT 5 versions, outperform in both indices. The Claude family's post-conventional scores range from 66 (Haiku 3.5, lowest) to 82 (Sonnet 3.7, highest), with N2 scores ranging from 52.73 to 74.98. In addition, this family consistently shows high post-conventional reasoning, with Sonnet models outperforming Opus and Haiku variants. For the Copilot family, post-conventional scores range from 56 (Think Deeper, lowest) to 74 (Quick Response, highest), with N2 scores of 57.58 and 63.83, respectively, which shows that the quick response model performs better overall. For the DeepSeek family, both models achieve the same post-conventional score of 66, with N2 scores of 62.67 (V3) and 64.75 (Deep Think R1), which indicate minimal performance differences between variants.

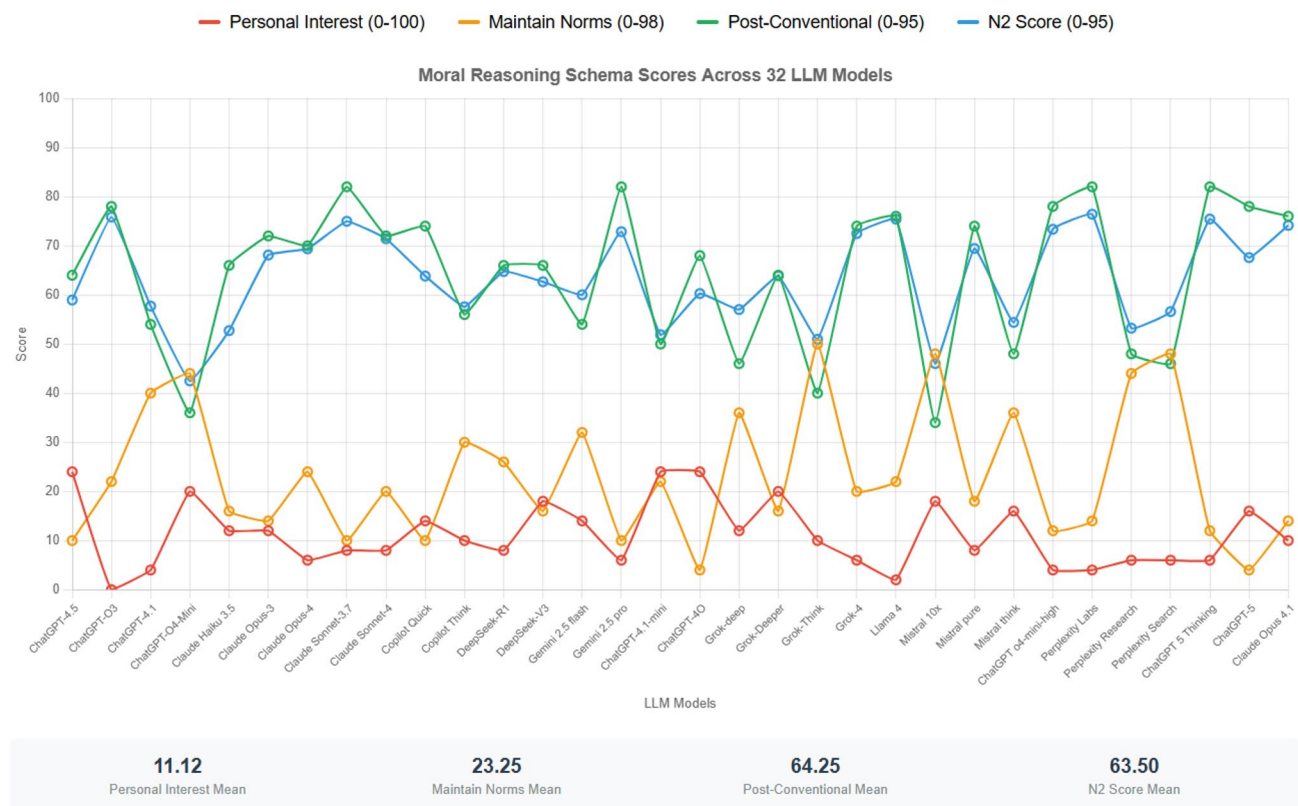


Fig. 2 LLM moral reasoning schema scores by model initial responses

Furthermore, for the Gemini Family, the Post-Conventional scores range from 54 (2.5 flash, lowest) to 82 (2.5 pro, highest), with N2 scores of 60.02 and 72.86, respectively, which indicate the pro model's clear advantage. The Grok family Post-Conventional scores range from 40 (Think, lowest) to 74 (Grok-4, highest), with N2 scores ranging from 50.97 to 72.48, which shows that newer versions outperform thinking variants. In addition, for the Mistral family, the Post-Conventional scores range from 34 (10x speed, lowest) to 74 (pure thinking, highest), with N2 scores ranging from 46.02 to 69.45, showing that pure thinking produces the best results. Finally, for the Perplexity family, Post-Conventional scores range from 46 (Search, lowest) to 82 (Labs, highest), with N2 scores ranging from 53.22 to 76.42, with Labs outperforming the others. Analysis of variance (ANOVA) was used to compare moral reasoning responses across eight LLM groups. Results revealed no statistically significant differences for any of the four moral reasoning variables: Personal Interest ($F(7, 23)=0.637$, $p=.721$), Maintaining Norms ($F(7, 23)=1.168$, $p=.359$), Post Conventional ($F(7, 23)=0.851$, $p=.558$), and N2 score ($F(7, 23)=0.500$, $p=.825$). However, the effect sizes revealed important information about practical significance. The Maintaining Norms variable had a large effect size ($\eta^2=0.262$), which indicates that although not statistically significant, there may be practical differences in how different LLM families approach Maintaining Norms reasoning. Post-Conventional reasoning and Personal Interest reasoning had large effect sizes ($\eta^2=0.206$ and 0.162 , respectively), whereas the N2 Score had a medium effect ($\eta^2=0.132$).

8.2 Comparison between LLM and human on moral reasoning

The analysis compared moral reasoning patterns between LLM responses ($n=32$) and human responses ($n=73,740$). The descriptive statistics revealed substantial differences in LLM and human moral reasoning patterns. LLMs showed significantly higher post-conventional reasoning than humans ($M=64.25$ vs. $M=34.86$). In addition, LLMs had significantly higher N2 scores than humans ($M=63.50$ vs. $M=34.26$). In comparison, LLMs demonstrated lower personal interest reasoning ($M=11.12$ vs. human $M=25.77$) and maintaining norms reasoning ($M=23.25$ vs. human $M=33.79$).

In addition, Levene's test for equality of variances revealed that the assumption of homogeneity of variances was violated for Personal Interest ($p < .001$) and N2 Score ($p = .001$) variables, whereas it was met for Post Conventional ($p = .531$) and Maintaining Norms ($p = .858$) variables. Given the violations of homogeneity of variance assumptions, Mann-Whitney U tests were used as the main analysis method, while independent samples t-tests were performed for comparative purposes (Table 3).

Non-parametric Mann-Whitney U tests found significant differences between LLM and human responses for all moral reasoning variables (all $p < .001$). Effect sizes were consistently large, with LLMs showing significantly higher post-conventional reasoning ($Z = -7.893$) and N2 scores ($Z = -8.640$) than humans, but significantly lower personal interest reasoning ($Z = -6.708$) and maintaining norms reasoning ($Z = -4.110$). Independent samples t-tests confirmed these findings. The t-test results showed significant findings for personal interest ($t(73766) = -6.469$, $p < .001$), maintaining norms ($t(73766) = -4.239$, $p < .001$), and post conventional (P score) ($t(73766) = 10.276$, $p < .001$). Additionally, the N2 score result was reported as $t(73737) = 29.237$, $p < .001$. Both parametric (t-tests) and non-parametric (Mann-Whitney U) methods showed identical results (all $p < .001$), which provides strong evidence for observed differences despite violations of homogeneity of variance assumptions. The consistency of these approaches increases confidence in the results. The analyses revealed significant effect sizes across all moral reasoning dimensions. Post-conventional reasoning had a large effect (Cohen's $d=1.817$), which shows that LLMs have significantly higher levels of post-conventional moral reasoning than humans. The N2 score difference had the largest effect size (Cohen's $d=1.837$), indicating that LLMs have a superior overall moral development profile. Personal interest reasoning had a significant negative effect (Cohen's $d = -1.144$), indicating that LLMs are much less likely to make moral decisions based on self-interest. In addition, Maintaining Norms had a medium effect, with Cohen's $d = -0.750$. Furthermore, the large effect sizes observed show that these differences are not only statistically significant but also represent meaningful and practically significant distinctions in moral reasoning patterns.

Table 3 Mann-Whitney U test results

Variable	Mann-Whitney U	Z	p	LLM mean rank	Human mean rank	Direction
Personal interest	372,697.50	-6.708	<0.001***	11,663.30	36,895.45	LLM < Human
Maintaining norms	685,239.00	-4.110	<0.001***	21,430.22	36,891.21	LLM < Human
Post conventional	229,683.50	-7.893	<0.001***	66,574.89	36,871.61	LLM > Human
N2 score	139,121.00	-8.640	<0.001***	69,375.97	36,855.89	LLM > Human

*** $p < .001$

8.3 Prompt engineering effects: initial vs. unbiased responses

In this section, the schema distributions and N2 scores for 32 LLMs in response to unbiased prompts are presented. The results are shown in Table 4.

Under unbiased prompting conditions, personal interest reasoning showed a compressed range (0–18) compared to initial responses (0–24). Several models achieved the minimum score of 0 (ChatGPT o3, Claude Sonnet 4, Llama 4, Perplexity Labs, Perplexity Research, and Claude Opus 4.1). This indicates that explicit bias reduction instructions successfully guided LLMs away from self-centered moral considerations, as the maximum score decreased from 24 to 18.

The improvements to the Maintaining Norms Schema resulted in a significant shift towards higher conventional

reasoning, with scores ranging from 8 to 50, up from 4 to 50 initially. In particular, the lower bound increased from 4 to 8, and several models received higher scores (ChatGPT-o4-mini: 50, ChatGPT-4.1-mini: 50, Grok-Think: 50). This suggests that unbiased prompting caused LLMs to rely more heavily on established social norms and moral frameworks.

Post-conventional reasoning had a similar range (34–82) as initial responses (34–82). Individual model performance, however, showed significant shifts: some models, such as Perplexity Research, achieved the maximum score of 80, while others, such as Mistral 10x speed, dropped to 34. The preservation of the broad range suggests that higher-level Post-conventional reasoning was resilient to unbiased prompting, even though individual models responded differently.

The N2 scores had a similar range (46.02–78.28) as the initial responses (42.48–76.42), which indicates that overall moral development profiles remained stable under unbiased conditions. The slight upward shift in both bounds (46.02 vs. 42.48 minimum, 76.47 vs. 76.42 maximum) suggests that unbiased prompting may have improved moral reasoning patterns' coherence while preserving model diversity.

Several models showed significant changes under unbiased conditions: Copilot Think Deeper increased maintaining norms from 30 to 38 while decreasing post-conventional from 56 to 38, and Claude Sonnet 4 increased maintaining norms from 20 to 46 while decreasing post-conventional from 72 to 54. These patterns indicate that some models interpreted “unbiased” instructions as requiring stricter adherence to conventional social norms rather than abstract principles.

In summary, the result shows that unbiased prompting instructions impacted lower-level moral reasoning schemas (personal interest and maintaining norms) while retaining higher-level post-conventional reasoning capabilities. The instruction to be “unbiased” appears to have been interpreted by LLMs as reducing self-interested considerations and increasing reliance on established social conventions, which indicates a shift toward more conventional moral reasoning approaches rather than purely principled ones (Fig. 3).

To determine whether the prompt modification influenced LLM moral reasoning responses, paired samples t-tests were performed on the initial ($n=32$) and unbiased responses ($n=32$) across four key measures: post-conventional reasoning scores (Table 5), Personal Interest scores, Maintaining Norms scores, and N2 scores. The analysis included 64 LLM models in total (32 per group).

The descriptive statistics show that personal interest reasoning decreased under unbiased prompting (11.12 vs. 7.25), whereas maintaining norms reasoning increased (23.25 vs. 30.06). Post-conventional reasoning differed little between

Table 4 Schema distribution and N2 scores across LLMs for unbiased responses

LLMs	Personal interest	Maintaining norms	<i>P</i> score	N2score
ChatGPT- 4.5	12	20	68	66.39
ChatGPT o3	0	24	76	76.18
ChatGPT-4.1	4	40	54	57.81
ChatGPT-o4-mini	2	50	44	53.60
Claude Haiku 3.5	12	26	56	52.50
Claude Opus 3	6	20	74	72.60
Claude Opus 4	14	8	78	74.11
Claude Sonnet 3.7	4	18	78	75.43
Claude Sonnet 4	0	46	54	60.78
Copilot quick response	14	22	64	61.70
Copilot Think deeper	16	38	38	47.00
DeepSeek-Deep Think R1	2	36	62	65.12
DeepSeek-V3	16	34	50	56.15
Gemini 2.5 flash	10	36	52	59.17
Gemini 2.5 pro	2	28	70	68.58
GPT-4.1-mini	6	50	42	55.48
GPT-4O	10	16	70	66.82
Grok – Deep Search	12	36	46	57.02
Grok – Think	10	50	40	50.97
Grok – Deeper Search	12	44	44	54.99
Grok-4	6	20	74	71.75
Llama 4	0	24	76	75.41
Mistral 10x speed	18	48	34	46.02
Mistral pure thinking	8	18	74	69.45
Mistral think	16	36	48	54.40
ChatGPT o4-mini – high	4	12	78	73.36
Perplexity Labs	0	34	66	67.96
Perplexity Research	0	18	80	76.47
Perplexity Search	6	48	46	56.60
ChatGPT 5 Thinking	6	12	82	76.05
ChatGPT-5	4	30	66	68.05
Claude Opus 4.1	0	20	80	78.28

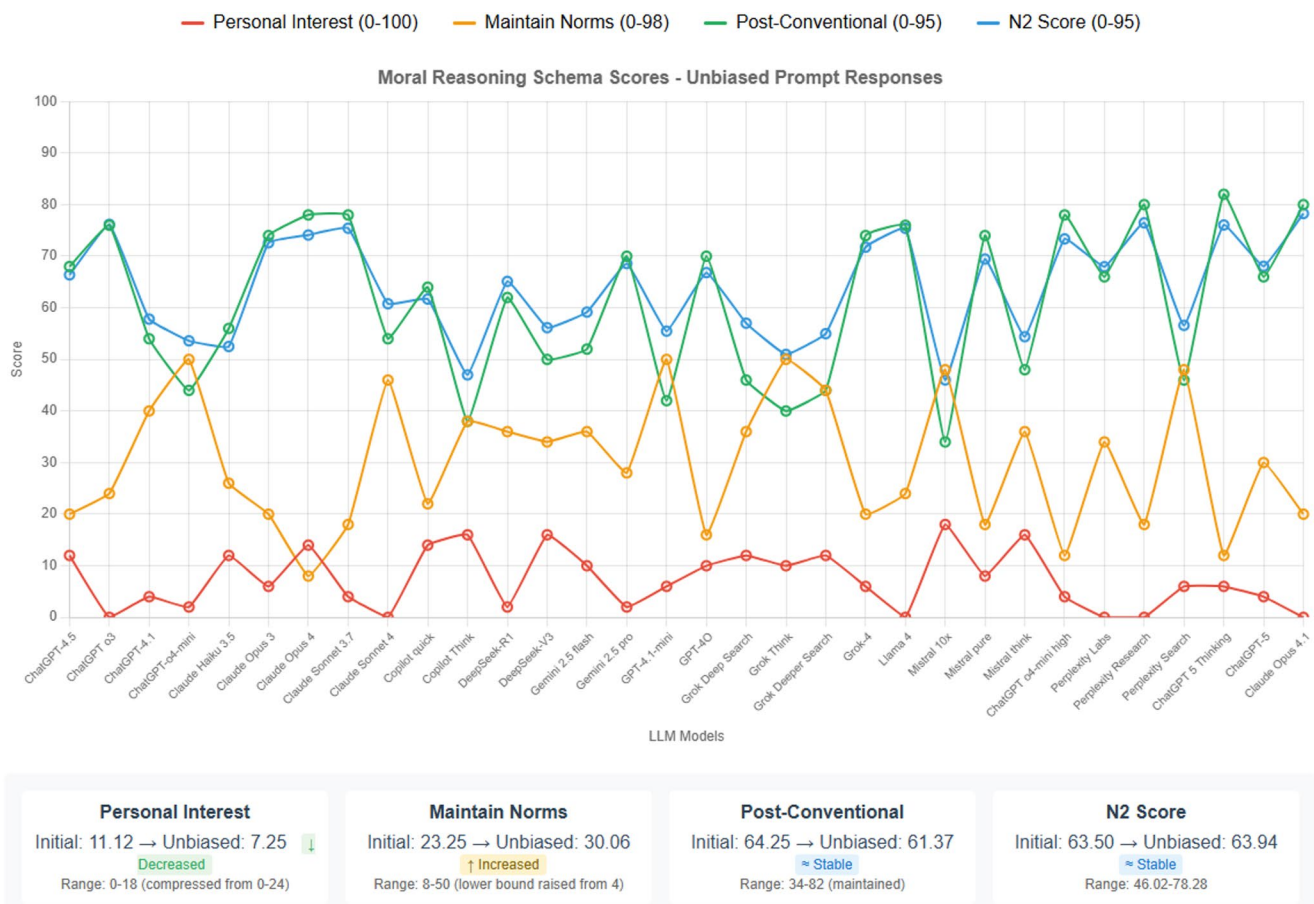


Fig. 3 LLM moral reasoning schema scores by model, for unbiased responses

Table 5 Paired samples T-test results

Measure	t	df	P	MD	95% CI	Cohen's d	d 95% CI
Personal interest	3.524	31	0.001*	3.875	[1.63, 6.12]	0.623	[0.24, 1.00]
Maintaining norms	-3.140	31	0.004*	-6.812	[-11.24, -2.39]	-0.555	[-0.92, -0.18]
Post-conventional	1.583	31	0.124	2.875	[-0.83, 6.58]	0.280	[-0.08, 0.63]
N2 score	-0.380	31	0.707	-0.447	[-2.85, 1.95]	-0.067	[-0.41, 0.28]

* $p < .05$, † $p < .10$

conditions (64.25 vs. 61.37). N2 scores remained nearly constant (63.50 vs. 63.94), which indicates that overall moral development profiles remained stable despite prompt changes. Levene's test for equality of variances indicated that the assumption of equal variances was met for all variables (all $p > .05$): Personal Interest ($p = .447$), Maintaining Norms ($p = .884$), Post-Conventional ($p = .570$), and N2 score ($p = .783$).

The paired samples t-test results demonstrated significant differences for Personal Interest ($p = .001$, Cohen's $d = 0.623$) and Maintaining Norms ($p = .004$, Cohen's $d = -0.555$) between initial and post-intervention conditions, with moderate effect sizes. However, post-conventional reasoning ($p = .124$, Cohen's $d = 0.280$) and the N2 score ($p = .707$, Cohen's $d = -0.067$) showed no significant differences.

This finding indicates that post-conventional reasoning and N2 scores were unaffected by the prompt modification, which suggests that these higher-level reasoning processes are more resistant to prompt variations. These findings indicate that the prompt design can influence LLM moral reasoning, but the effects are limited and focused on lower-level moral reasoning schemas rather than sophisticated moral reasoning.

8.4 Intervention analysis: historical figure influences

The intervention analysis looked at how introducing historical moral figures influenced LLM reasoning patterns in five moral dilemmas. Two prompt types were used: Prompt

A provided clear introductions to influential figures' principles with guiding questions, and Prompt B replaced original characters with historical figures, allowing for direct persona adoption (see Interventions section for detailed descriptions). The schema distributions and N2 scores for 32 LLMs in response to intervention prompts are presented in Table 6.

The results in Table 6 show the effect of historical moral figure interventions on LLM moral reasoning patterns in five ethical dilemmas. The findings show a consistent shift from post-conventional to conventional moral reasoning when LLMs were exposed to historical moral figures using both intervention approaches (Fig. 4).

Personal interest reasoning remained consistent across conditions, with minimal variation in ranges for initial responses (0–24), Prompt A (0–20), and Prompt B (2–24). Eventually, this consistency suggests that interventions primarily influenced higher-level moral reasoning schemas rather than self-centered considerations.

Both intervention prompts resulted in significant improvements in maintaining norms reasoning compared to initial responses (range: 4–50). Prompt A interventions ranged from 18 to 50, while Prompt B interventions ranged from 18 to 48, which indicates an increase in the lower range from 4 to 18. It means that historical moral figures influenced LLMs to prioritize established social norms, cultural traditions, and conventional moral expectations.

Post-conventional Reasoning Patterns: Compared to initial responses (range: 34–82), both intervention conditions showed significant decreases in post-conventional reasoning: Prompt A interventions ranged from 34 to 74, and Prompt B interventions ranged from 34 to 74. This represents a narrowing of the upper range, which indicates that historical moral figure guidance successfully shifted LLMs away from the most abstract universal principles and toward more contextual, conventional approaches.

N2 Score: The comprehensive N2 scores narrowed from initial responses (range: 42.48–76.42) to both intervention conditions (Prompt A: range: 42.10–74.90; Prompt B: range: 39.21–74.41), which shows a general shift toward more conventional moral reasoning patterns while maintaining coherent moral development profiles.

8.4.1 One-way ANOVA results

A one-way ANOVA was conducted to compare the initial responses between Prompt A and B interventions. The descriptive statistics for the three groups are presented in Table 7.

Levene's test was conducted to assess the homogeneity of variance assumption for each dependent variable. The results indicated that the assumption of equal variances

was met for three variables ($p > .05$): Personal Interest ($p = .499$), post-conventional ($p = .180$), and N2 score ($p = .417$). However, the assumption was violated for Maintaining Norms ($p = .042$). One-way ANOVAs were conducted to examine differences between LLM groups on each moral reasoning variable. The results are summarized in Table 8.

The ANOVA analysis reveals significant main effects for three of the four variables, which shows that historical moral figure interventions effectively modified LLM moral reasoning patterns. For personal interest schema score, the ANOVA showed no significant effect ($p = .449$, $\eta^2 = 0.017$), indicating that historical figure interventions did not significantly impact personal interest moral considerations. The negligible effect size confirms that personal interest reasoning remained consistent across all conditions, which suggests that interventions focused on higher-level moral reasoning schemas rather than basic personal interest reasoning.

The Maintaining Norms Schema showed a significant effect ($p = .001$, $\eta^2 = 0.174$), with a large effect size indicating practical significance. This demonstrates that historical moral figure guidance effectively increased LLMs' reliance on conventional social rules, cultural traditions, and established moral frameworks, which indicates a fundamental shift toward more traditional moral reasoning approaches.

The Post-Conventional Schema analysis revealed a significant effect ($p < .001$, $\eta^2 = 0.149$) with a large effect size, which indicates significant practical impact. This suggests that historical figure interventions effectively reduced LLMs' reliance on abstract universal principles, shifting them away from the highest levels of moral reasoning and toward more contextual, conventional approaches.

The N2 Score showed a significant effect ($p = .002$, $\eta^2 = 0.130$), which indicates significant changes in overall moral development profiles. The N2 measure captures both the decrease in post-conventional reasoning and the increase in conventional approaches, which confirms that interventions resulted in coherent shifts in moral reasoning patterns while remaining developmentally consistent.

Welch's robust tests were used to address potential violations of homogeneity assumptions. As mentioned before, Levene's tests indicated that the Maintaining Norms variable violated the homogeneity assumption ($p = .042$), which makes the Welch test more appropriate for this analysis. The Welch tests confirmed all ANOVA findings: Personal Interest remained non-significant ($p = .424$), while Maintaining Norms ($p = .001$), Post-Conventional ($p = .002$), and N2 Score ($p = .002$) all maintained significance. The convergence of standard ANOVA and robust Welch tests provides strong support for the results.

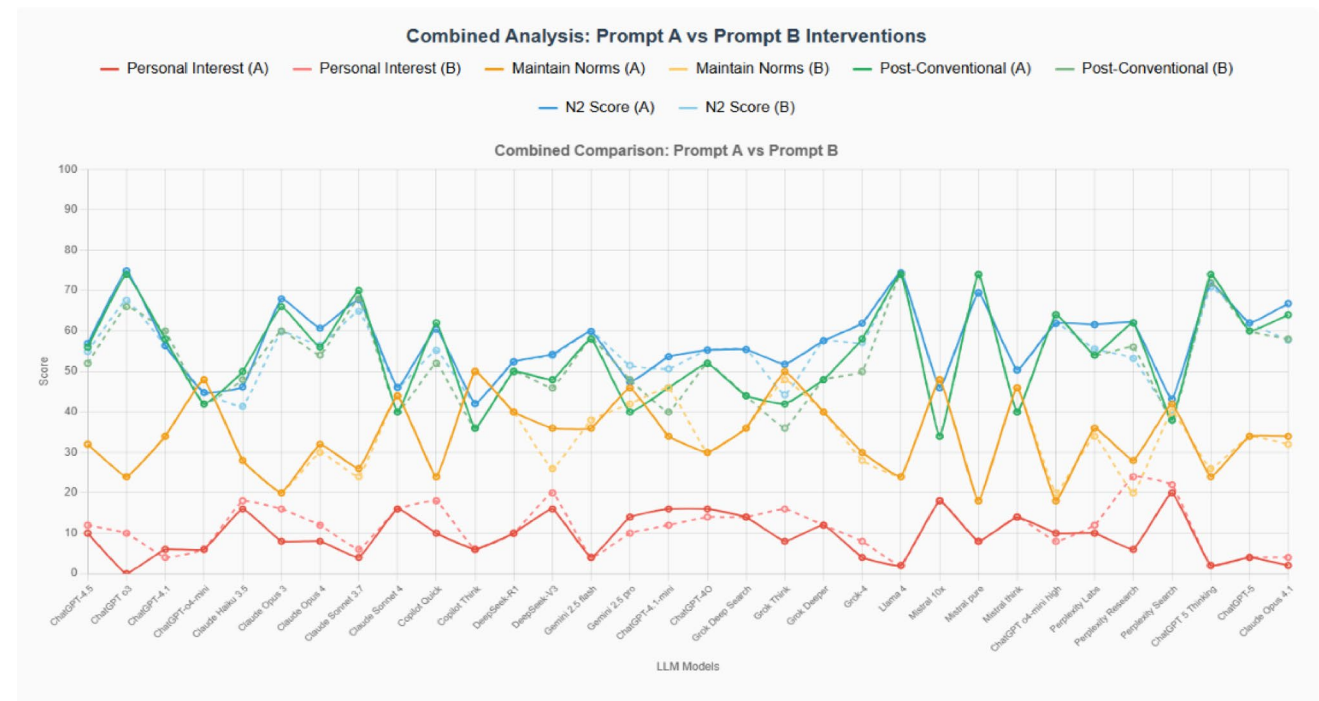
For group comparison, the Tukey test was chosen because equal variances were assumed for most variables (Levene's

Table 6 Schema distribution and N2 scores across LLMs for interventions

ID	Personal interest	Maintaining norms	P score	N2 score
ChatGPT – 4.5 Prompt A	10	32	56	56.89
ChatGPT – 4.5 Prompt B	12	32	52	54.89
ChatGPT o3 – Prompt A	0	24	74	74.90
ChatGPT o3 – Prompt B	10	24	66	67.58
ChatGPT-4.1 – Prompt A	6	34	58	56.32
ChatGPT-4.1 – Prompt B	4	34	60	56.39
ChatGPT-o4-mini–Prompt A	6	48	42	44.85
ChatGPT-o4-mini–Prompt B	6	48	42	44.85
Claude Haiku 3.5 Prompt A	16	28	50	46.24
Claude Haiku 3.5 Prompt B	18	28	48	41.40
Claude Opus 3 – Prompt A	8	20	66	67.96
Claude Opus 3 – Prompt B	16	20	60	59.76
Claude Opus 4 – Prompt A	8	32	56	60.69
Claude Opus 4 – Prompt B	12	30	54	56.47
Claude Sonnet 3.7 – Prompt A	4	26	70	67.72
Claude Sonnet 3.7 – Prompt B	6	24	68	64.85
Claude Sonnet 4 – Prompt A	16	44	40	46.11
Claude Sonnet 4 – Prompt B	16	44	40	46.11
Copilot quick response– Prompt A	10	24	62	60.44
Copilot quick response– Prompt B	18	24	52	55.17
Copilot Think Deeper – Prompt A	6	50	36	42.10
Copilot Think Deeper – Prompt B	6	50	36	42.10
DeepSeek-Deep Think R1 – Prompt A	10	40	50	52.36
DeepSeek-Deep Think R 1– Prompt B	10	40	50	52.36
DeepSeek-V3 Prompt B	20	26	46	53.92
DeepSeek-V3 Prompt A	16	36	48	54.21
Gemini 2.5 flash Prompt A	4	36	58	59.89
Gemini 2.5 flash Prompt B	4	38	58	59.79
Gemini 2.5 pro Prompt A	14	46	40	47.47
Gemini 2.5 pro Prompt B	10	42	48	51.47
ChatGPT – 4.1-mini – Prompt A	16	34	46	53.63
ChatGPT – 4.1-mini – Prompt B	12	46	40	50.63
ChatGPT – 4O – Prompt A	16	30	52	55.25
ChatGPT – 4O – Prompt B	14	30	52	55.25
Grok – Deep Search – Prompt A	14	36	44	55.39
Grok – Deep Search – Prompt B	14	36	44	55.39
Grok – Think Prompt A	8	50	42	51.75
Grok – Think Prompt B	16	48	36	44.26
Grok – Deeper Search – A	12	40	48	57.56
Grok – Deeper Search – Prompt B	12	40	48	57.56
Grok-4 Prompt A	4	30	58	61.91
Grok-4 Prompt B	8	28	50	57.11
Llama 4 Prompt A	2	24	74	74.41
Llama 4 Prompt B	2	24	74	74.41
Mistral 10x speed – Prompt A	18	48	34	46.02
Mistral 10x speed – Prompt B	18	48	34	46.02
Mistral pure thinking– Prompt A	8	18	74	69.45
Mistral pure thinking– Prompt B	8	18	74	69.45
Mistral think– Prompt A	14	46	40	50.39
Mistral think– Prompt B	14	46	40	50.39
ChatGPT o4-mini – high - Prompt B	8	20	64	61.88
ChatGPT o4-mini –high- Prompt A	10	18	64	61.88
Perplexity Labs – Prompt A	10	36	54	61.57
Perplexity Labs – Prompt B	12	34	54	55.53
Perplexity Research – Prompt A	6	28	62	62.02

Table 6 (continued)

ID	Personal interest	Maintaining norms	P score	N2 score
Perplexity Research – Prompt B	24	20	56	53.22
Perplexity Search – Prompt A	20	42	38	43.22
Perplexity Search – Prompt B	22	40	38	39.21
ChatGPT 5 Thinking – Prompt A	2	24	74	71.81
ChatGPT 5 Thinking – Prompt B	2	26	72	70.80
ChatGPT-5 – Prompt A	4	34	60	61.99
ChatGPT-5 – Prompt B	4	34	60	61.99
Claude Opus 4.1 Prompt A	2	34	64	66.79
Claude Opus 4.1 Prompt B	4	32	58	57.72

**Fig. 4** LLM moral reasoning - historical figure intervention analysis**Table 7** Descriptive statistics for all groups

Measure	Group	N	Mean	SD	SE	95% CI lower	95% CI upper
Personal interest	Initial	32	11.12	6.66	1.17	8.72	13.52
	Prompt A	32	9.50	5.58	0.987	7.48	11.51
	Prompt B	32	11.18	5.76	1.01	9.10	13.26
Maintaining norms	Initial	32	23.25	13.56	2.39	18.36	28.13
	Prompt A	32	33.81	9.48	1.67	30.39	37.23
	Prompt B	32	33.87	9.58	1.69	30.41	37.33
Post-conventional (P score)	Initial	32	64.25	14.52	2.56	59.01	69.48
	Prompt A	32	54.12	12.14	2.14	49.74	58.50
	Prompt B	32	52.37	11.28	1.99	48.30	56.44
N2 score	Initial	32	63.50	9.55	1.68	60.05	66.94
	Prompt A	32	57.59	9.14	1.61	54.29	60.89
	Prompt B	32	55.26	8.60	1.52	52.15	58.36

Table 8 One-way ANOVA and robust tests (Welch) results

Variable	Standard ANOVA				Welch test (Robust)			
	F	df	<i>p</i>	η^2	F	df1	df2	<i>p</i>
Personal interest	0.807	2, 93	0.449	0.017	0.870	2	61.676	0.424
Maintaining norms	9.821	2, 93	<0.001	0.174	7.743	2	60.748	0.001
Post conventional (P score)	8.119	2, 93	<0.001	0.149	7.112	2	61.386	0.002
N2 score	6.951	2, 93	0.002	0.130	6.730	2	61.883	0.002

Table 9 Additional DIT-2 indices across experimental conditions

Index	Range	Baseline	Unbiased	Intervention
ASCORE	0–16	0–3 (M=0.59)	0–4 (M=0.56)	0–7 (M=1.14)
(Antisocial)				
MSCORE	0–16	0–3 (M=0.09)	0–3 (M=0.09)	0–3 (M=0.14)
(Meaningless)				
U-Score (Utilizer)	–1 to 1	–0.13 to 0.27 (M=0.09)	–0.15 to 0.19 (M=0.02)	–0.12 to 0.30 (M=0.16)
CANCER10 (Religious Orthodoxy)	1–9	1–7 (M=2.93)	2–5 (M=3.06)	2–9 (M=6.9)
NUMCD (Cannot Decide)	0–5	0–4 (M=0.6)	0–5 (M=3.31)	0–5 (M=0.3)
HUMLIB (Humanitarian Liberalism)	0–5	1–5 (M=3.4)	0–4 (M=1.03)	0–4 (M=2.7)
CONSTRAN (Consolidation)	1–2	2 (100%)	2 (100%)	1–2 (94% = 2)
TYPENEW (Type Indicator)	1–7	Type 4: 12% Type 7: 88%	Type 4: 16% Type 7: 84%	Type 4: 28% Type 5–6: 6% Type 7: 66%

N=32 LLMs tested across three conditions. no models exceeded purging thresholds. Percentages for TYPENEW represent the proportion of models in each category. M=mean across all models in conditions

test $p > .05$), and it was necessary to control for the family-wise error rate across multiple pairwise comparisons. However, for Maintaining Norms, which violated homogeneity assumptions ($p = .042$), Games-Howell post-hoc tests were also conducted to provide robust comparisons.

For maintaining norms, Initial (baseline) responses Tukey HSD showed significant differences (–10.56 vs. Prompt A, $p = .001$; –10.63 vs. Prompt B, $p = .001$) with no difference between Prompt A and Prompt B ($p = 1.000$). Due to the violation of homogeneity assumptions for Maintaining Norms, a Games-Howell post-hoc comparison was conducted, which confirmed the Tukey HSD results. In addition, Tukey HSD showed that post-conventional reasoning decreased from Initial (10.13 vs. Prompt A, $p = .006$; 11.88 vs. Prompt B, $p = .001$), with equivalent effects between methods ($p = .847$). N2 scores decreased from Initial (5.91 vs. Prompt A, $p = .030$; 8.24 vs. Prompt B, $p = .001$) with similar reductions across prompt types ($p = .563$).

Post-hoc analyses using Tukey HSD and Games-Howell (for Maintaining Norms) show that both intervention types had statistically equivalent effects, with no significant differences between the Prompt A and Prompt B conditions

across any variable. This suggests that both approaches to introducing historical moral figures effectively shifted LLM responses away from higher-level post-conventional reasoning and toward more conventional, rule-based moral reasoning.

8.5 Exploratory analysis

An exploratory analysis was also conducted on additional DIT-2 indices across experimental conditions to identify potentially useful patterns beyond the primary moral reasoning scores, with results presented in Table 9.

Validity indicators confirmed high task engagement under all experimental conditions. First, the antisocial scores (ASCORE) were consistently low (0–7), which shows minimal anti-establishment attitudes and absence of stage 4½ reasoning patterns that criticize authorities for hypocrisy. Second, meaningless item scores (MSCORE) were constantly low (0–3), below the purging threshold of 10, which indicates that LLMs didn't support meaningless items and engaged in valid moral reasoning tasks rather than fabricating highly complex responses. Finally, these patterns remained consistent across baseline, unbiased, and intervention conditions, which suggests that prompt conditions did not affect response integrity.

Utilizer scores (U) were generally positive across conditions (–0.13 to 0.30). These scores indicate a reasonable consistency between the moral principles rated as important by models and their action choices in dilemmas. While most models demonstrated consistent action choice-item ratings alignment, some negative scores indicated that some models had difficulty translating abstract moral commitments into concrete behavioral decisions, especially under unbiased conditions.

Three indices demonstrated significant sensitivity to experimental manipulations. First, religious orthodoxy (CANCER10) scores more than doubled during intervention conditions ($M=6.9$) compared to baseline ($M=2.93$). This is important, particularly as historical religious figures like Pope John Paul II were included. When authoritative moral exemplars guided the story (story 4), LLMs successfully incorporated traditional religious and spiritual considerations into end-of-life moral reasoning, as demonstrated by this dramatic increase. Second, humanitarian liberalism

Table 10 One-way ANOVA and (Welch) tests results

Variable	Standard ANOVA				Welch test (Robust)			
	F	df	<i>p</i>	η^2	F	df1	df2	<i>p</i>
ASCORE	1.947	2, 93	0.149	0.040	2.381	2	58.097	0.101
MSCORE	0.095	2, 93	0.909	0.002	0.092	2	61.709	0.912
Utilizer Score	5.156	2, 93	0.008*	0.100	4.821	2	61.775	0.011*
Religious Orthodoxy	40.539	2, 93	<0.001*	0.466	61.650	2	54.153	<0.001*
Cannot Decide Choices	2.744	2, 93	0.070†	0.056	4.713	2	48.700	0.013*
Humanitarian Liberalism	9.164	2, 93	<0.001*	0.165	7.457	2	59.934	0.001*

* $p < .05$, † $p < .10$. η^2 = Eta-squared (effect size)

(HUMLIB) scores decreased significantly after unbiased prompting ($M=1.03$ vs. baseline $M=3.4$), indicating that explicit instructions to produce “unbiased” responses were interpreted as directives to avoid politically progressive positions and adopt more neutral views. Third, under unbiased conditions, the can-not-decide choices (NUMCD) increased significantly ($M=3.8$, with 50% of models selecting the maximum of 5) compared to the baseline ($M=0.6$), which suggests improved deliberative processing and recognition of moral complexity when instructed to avoid bias.

The Type Indicator (TYPENEW) revealed that most models maintained consolidated post-conventional (Type 7) profiles across conditions, while intervention conditions resulted in a slight increase in maintaining norms (Type 4) profiles from 12% at baseline to 28%. In addition, intervention conditions produced the only instances of transitional profiles (Types 5–6), as seen in Claude Haiku 3.5 and Copilot Think Deeper, which indicates that historical moral figure guidance caused developmental disequilibrium in certain models. Consolidation scores (CONSTRAN) remained primarily at 2 (indicating consolidated profiles) across 94–100% of models under all conditions (except Claude Haiku 3.5 and Copilot Think Deeper under intervention conditions), which confirms that prompt manipulations did not destabilize moral reasoning schemas despite shifting specific moral considerations.

8.5.1 Unbiased prompt condition for additional indices

A paired sample *t*-test was used to compare additional indices between initial and unbiased responses. Results showed three significant changes. Cannot Decide Choices increased dramatically ($t(31) = -7.830$, $p < .001$), which indicates substantially greater indecisiveness at the unbiased prompt condition. Humanitarian Liberalism decreased significantly ($t(31)=7.736$, $p < .001$), suggesting variable individual change. Utilizer Score showed a significant decrease ($t(31)=2.945$, $p = .006$), which shows lower consistency of moral reasoning and action choices. The remaining measures, ASCORE ($t(31)=0.254$, $p=.801$) and Religious Orthodoxy ($t(31) = -1.072$, $p=.292$), showed no significant changes, which indicates stability. MSCORE was excluded

from analysis as only one model scored non-zero (Grok Deep Search score=3), while all other 31 models scored zero. The zero score shows that LLMs were not trapped with meaningless items, which shows engagement in the moral reasoning task rather than response fabrication.

8.5.2 One-way ANOVA results for additional indices

A one-way ANOVA was used to compare the additional DIT-2 indices in the initial and intervention groups. Some variables (ASCORE, Religious Orthodoxy, Cannot Decide Choices, Humanitarian Liberalism) violated the homogeneity assumptions, so robust Welch tests were used in addition to standard ANOVA. Significant differences were found for Utilizer Score, Religious Orthodoxy, Cannot Decide Choices, and Humanitarian Liberalism. Religious Orthodoxy had the strongest effect ($\eta^2=0.466$), while Humanitarian Liberalism ($\eta^2=0.165$) and Utilizer Scores ($\eta^2=0.100$) were moderately responsive to prompts (Table 10).

Tukey HSD Post-hoc comparisons test shows that Prompt B significantly improved utilizer scores compared to the initial condition ($p = .008$), while Prompt A showed a non-significant trend ($p = .056$). The two prompts did not differ significantly ($p = .742$). The Games-Howell post-hoc comparisons showed both intervention prompts (A and B) significantly increased religious orthodoxy scores compared to initial responses ($p < .001$), and there was no significant difference between the two intervention conditions. For humanitarian liberalism, initial conditions had significantly higher scores than both intervention responses, which shows that historical figure guidance decreased consideration of humanitarian values slightly. Also, for “cannot decide” choices, the initial and prompt B groups differed significantly, which shows that the initial condition had slightly more cannot decide choices. In conclusion, this exploratory analysis revealed that LLMs revealed distinct patterns in response to experimental conditions across some additional DIT-2 indices.

Eventually, to determine whether the intervention for Story 4, which featured the historical religious figure Pope John Paul II, had a unique influence on the observed patterns in moral reasoning, a comparative analysis was performed

with and without this dilemma. The analysis revealed that the directional patterns were consistent across the four- and five-story analyses.

9 Discussion

9.1 Overview of findings

This study provides robust evidence that LLMs have sophisticated moral reasoning capabilities in the domain of bedrock or non-contextual reasoning that are significantly different from human patterns. The results reveal a few key insights: LLMs show significantly higher levels of post-conventional moral reasoning than humans, rely less on personal interest and conventional norm-based reasoning, and demonstrate adaptability to prompt engineering and contextual interventions.

9.2 LLM moral reasoning profiles

The most significant result is that LLM models consistently have higher post-conventional reasoning scores, with a mean of 64.25 (N2 score 63.50) compared to human averages of 34.86 (N2 score 34.26). In addition, the high N2 scores support this interpretation, as this measure specifically captures both postconventional prioritization and the rejection of personal-interest items. Based on the neo-Kohlbergian theory, these findings highlight fundamental differences in how LLMs and humans activate moral schemas. According to the theory, individuals use different schemas depending on the situation, with more developmentally advanced individuals scoring higher on post-conventional reasoning [19, 46]. In addition, LLMs' consistent preference for post-conventional schemas ($M = 64.25$) over maintaining norms ($M = 24.62$) shows that they have internalized principled ethical frameworks based on training data that prioritize universal principles over conventional social expectations. However, this pattern differs from typical human moral development, in which maintaining norms reasoning frequently functions as a critical developmental bridge between personal interests and post-conventional reasoning [42]. The low maintaining norms scores in LLMs may indicate an inadequate level of understanding of the social and cultural foundations that underpin mature moral reasoning in humans.

10 Human-AI moral reasoning comparisons

The significant differences between LLM and human moral reasoning patterns highlight fundamental differences in how AI systems and humans approach moral reasoning.

The results revealed that LLMs' significantly higher post-conventional reasoning scores and N2 score, combined with lower personal interest and maintaining norms scores, show a moral reasoning profile that prioritizes universal ethical principles while de-emphasizing both self-interested motivations and conventional rule-based reasoning.

This pattern is very different from typical human moral development, which involves the gradual emergence of post-conventional reasoning through life experience, education, and cognitive maturation [46]. Human moral reasoning is rooted in social relationships, personal identities, and cultural contexts, as evidenced by our reliance on personal interests and social conventions [7]. LLMs appear to rely on abstract principled reasoning patterns learned from philosophical and ethical texts [9]. In addition, the large effect sizes observed (Cohen's d ranging from -1.147 to 1.820) indicate that these are not statistical artifacts, but rather substantive differences in moral reasoning approaches.

10.1 Prompt engineering and contextual sensitivity

The selective effects of prompt modifications provide important information about the adaptability and robustness of LLM moral reasoning. The significant decrease in personal interest reasoning when "unbiased" prompts were used indicates that LLMs can be guided away from self-centered considerations through explicit instruction. This finding supports the possibility of using prompt engineering to align LLM outputs with desired ethical frameworks.

The significant increase in maintaining norms reasoning under unbiased conditions shows that reducing one type of bias may unintentionally increase reliance on traditional rule-based reasoning. This choice indicates that prompt engineering effects are not always positive and might require proper measurement to achieve the desired results.

The power of post-conventional reasoning and N2 scores to prompt modifications is especially noteworthy. This stability suggests that higher-level moral reasoning processes in LLMs are less prone to superficial prompt variations, which indicates solid internalization of principled reasoning approaches rather than surface-level pattern matching.

10.2 Historical figure interventions and adaptability

The intervention findings are one of the study's most important findings, which show that LLMs can systematically adapt their moral reasoning frameworks when presented with historical moral exemplars. In fact, the consistent shift from post-conventional reasoning (Initial $M = 64.25$) to more conventional approaches (Prompt A $M = 54.12$, Prompt B $M = 52.37$) across both intervention types shows

that LLMs can successfully incorporate contextual moral guidance and adjust their reasoning patterns as needed.

However, opposite to the previous research [20], the historic moral figures used may have activated the maintaining norms reasoning rather than post-conventional reasoning due to some possible reasons. These reasons include their association with established traditions, authority, and social orders in training corpora; temporal and cultural distance from contemporary contexts; representation in training data as exceptional rather than replication of moral figures; domain irrelevance; absence of autonomy-supportive engagement [22]; and the ceiling effect limits upward improvement from already-high baseline scores. Other than that, this finding is consistent with previous research [23], which found that historic figures are less effective than peer exemplars in promoting moral development, which suggests that historic figures might not be effective resources to promote moral development.

The ANOVA results confirm the effectiveness of these interventions, with large effect sizes for post-conventional reasoning ($\eta^2 = 0.149$), maintaining norms reasoning ($\eta^2 = 0.174$), and N2 scores ($\eta^2 = 0.130$). As a result, this adaptability has significant educational implications, which implies that LLMs could function as dynamic moral reasoning tools that show different ethical approaches based on historical or cultural contexts. The ability to change reasoning patterns while remaining consistent within each approach demonstrates sophisticated contextual understanding rather than simple pattern matching.

In addition, the similarity of Prompt A and Prompt B conditions suggests that the explicit introduction of moral figures and direct persona adoption have similar impacts. This versatility in intervention methods creates new opportunities for using moral guidance in LLM interactions. It also increases their potential value in educational and training settings.

The intervention effects are consistent with neo-Kohlbergian theory, which emphasizes contextual influences on schema activation [46]. The successful transition from post-conventional to maintaining norms reasoning when historical figures were introduced demonstrates that LLMs can change their schema preferences in response to contextual cues, which is a key feature of the neo-Kohlbergian model [46]. However, the ease and magnitude of these shifts (effect sizes $\eta^2 = 0.149$ – 0.174) raise questions about whether LLMs possess stable moral orientations that characterize mature human moral development or engage in contextual pattern matching without underlying moral understanding.

The exploratory analysis of additional DIT-2 indices revealed that LLM moral reasoning is far more complex and context-dependent than primary schema scores alone might suggest. The analysis of additional DIT-2 indices beyond

the primary moral reasoning scores revealed important patterns that enhance our understanding of how different prompt conditions influence LLM moral reasoning consistency, decisiveness, and value orientations. It presents comprehensive DIT-2 indices that provide deeper insights into the moral reasoning patterns and response validity of LLMs beyond the primary schema scores. In addition, the comparative analysis of moral reasoning patterns with and without Story 4 (the Pope John Paul II dilemma) provided significant methodological insights. Despite the fact that Story 4 contains explicit religious content and has significantly higher religious orthodoxy scores, excluding it from the analysis did not change the directional patterns observed across experiments. Both four- and five-story analyses revealed consistent decreases in post-conventional reasoning (P and N2) scores from baseline to intervention, as well as increases in Maintaining Norms reasoning. This convergence implies that the observed shifts in LLM moral reasoning were not the result of a single religiously themed dilemma, but rather of broader patterns in how these models respond to authoritative moral exemplar guidance. The robustness of these directional effects across different story designs is an interesting result that requires additional research to understand the mechanisms that influence moral reasoning patterns through in-context learning.

10.2.1 Alignment with previous research

The findings of this study are consistent with and extend previous research on LLM moral reasoning capabilities. The findings that LLMs have higher post-conventional reasoning scores than humans are consistent with our pilot study, which also found similar patterns [34], as well as higher scores than previous studies [20, 55]. For the interventions, while the findings differ from the previous study [20], they are in line with evidence that historic figures are less effective than peer exemplars for promoting moral development, and perhaps historic figures are not effective resources for promoting moral development toward post-conventional moral reasoning [23]. Overall, the consistency across DIT moral dilemmas and research groups increases confidence in the findings' validity and suggests that sophisticated moral reasoning patterns in LLMs are robust and replicable.

The finding that newer LLM models have higher moral reasoning scores is consistent with previous research on how LLMs' moral reasoning is affected by input data, training strategies, RLHF training effects, and model architectures [14, 48]. Larger models with improved RLHF training can follow sophisticated normative concepts and avoid harmful outputs when given clear ethical instructions [14]. This supports the findings that advanced models such as ChatGPT 5 Thinking and Gemini 2.5 pro outperformed earlier versions

across multiple moral reasoning measures. These findings across LLM versions provide strong empirical support for the conclusion that modern LLMs have developed sophisticated capabilities in fundamental moral reasoning, as measured by the DIT-2.

10.2.2 Implications for moral development and education

These findings have important implications for human moral education and development. The demonstrated ability of LLMs to demonstrate sophisticated moral reasoning patterns and adapt to various moral frameworks indicates that they could be used as educational tools to introduce students to advanced ethical frameworks and reasoning processes. However, because such tools differ from human moral reasoning patterns, careful consideration of how to incorporate them into moral education curricula is required. To develop mature moral judgment, students need to be provided with opportunities for dealing with real-world moral dilemmas involving personal stakes, emotional complexity, and genuine consequences. As a result, LLMs should be used to supplement rather than replace experiential moral learning, perhaps as tools for exploring various ethical perspectives while ensuring students engage genuinely with moral dilemmas that necessitate personal reflection, emotional processing, and practical decision-making in real-world settings.

The higher post-conventional reasoning demonstrated by LLMs could be a valuable supplement to human moral instruction by providing consistent exposure to principled reasoning approaches. Furthermore, students may benefit from seeing how universal ethical principles are applied in a variety of contexts, particularly in settings where human instructors may be influenced by personal biases or conventional reasoning.

Furthermore, the intervention outcomes hold particular promise for moral education. The ability of LLM reasoning patterns to shift in response to historical moral figures implies that these systems could be used as dynamic pedagogical tools to demonstrate how different moral traditions and exemplars approach moral dilemmas. This could introduce students to a variety of moral reasoning approaches while ensuring educational consistency.

11 Ethical implications and responsible development

The demonstrated moral reasoning capabilities of LLMs raise significant ethical concerns about their development and deployment. The ability to influence moral reasoning through prompt engineering and contextual interventions

emphasizes the responsibility of developers and users to ensure that these capabilities are used ethically.

The potential for LLMs to serve as moral advisors or educational tools necessitates careful consideration of their limitations and protective measures. While these systems indicate sophisticated reasoning patterns, LLMs lack consciousness [39], which requires human supervision and alternative perspectives [49]. The findings also raise concerns about the applicability of automated moral reasoning in critical situations. While current narrow AI systems can provide useful analytical support and alternative perspectives, assigning moral authority to AI systems raises serious ethical concerns. However, whether future Artificial General Intelligence (AGI) or Artificial Superintelligence (ASI) will develop capacities beyond the current narrow AI systems and more identical to human moral understanding remains an open question.

12 Limitations

Several limitations should be considered when interpreting these findings. First, the evaluation of LLM moral reasoning is based on text-based responses to hypothetical scenarios, which may not accurately reflect performance in real-world ethical situations with real consequences. The absence of personal interests, emotional engagement, and social pressure in these assessments may result in an overestimation of LLM's moral reasoning abilities.

Second, the higher moral reasoning scores seen in LLMs may be due to overfitting to philosophical and ethical content in training data rather than genuine moral understanding. Therefore, the statistical nature of LLM learning processes raises the question of whether observed patterns reflect genuine moral reasoning or sophisticated pattern matching based on textual associations. Third, the cultural and contextual reasoning underlying moral development makes it difficult to generalize these findings to different populations and societies, as all human data used to compare to LLM moral reasoning responses is from English-speaking participants in the United States. Future research should compare LLM responses with more diverse human data from different cultural contexts.

Finally, this study served as a preliminary test of whether exemplar-based interventions can effectively influence moral reasoning in LLMs using a diverse set of examples. However, we acknowledge as a limitation that different exemplars might produce different effect sizes, and the optimal choice of exemplar might vary with context or dilemma type.

12.1 Future directions

This study's findings provide important methodological foundations for assessing LLM moral reasoning capacities. First, adapting a recognized moral reasoning instrument (DIT-2) provides a standard tool for comparing LLM models and human populations. Second, the multidimensional approach, which takes into account group differences, human comparisons, prompt effects, and contextual interventions, offers a comprehensive assessment methodology that could inform future research in this field.

Third, the intervention methodology, which used both explicit guidance (Prompt A) and persona adoption (Prompt B) approaches, provides a solid foundation for studying LLM adaptability in moral reasoning contexts. The equivalence of these approaches suggests that researchers can design interventions while maintaining the integrity of their methodology. In addition, future research might systematically examine how moral exemplar characteristics (cultural background, specific moral focus, historical period) moderate intervention effects.

Fourth, for future studies, researchers should use larger samples (newer) of LLM models to detect meaningful differences between LLM groups and explore how moral reasoning generalizes across contexts and domains. In addition, future research could explore the potential benefits of bootstrapping resampling by presenting each DIT-2 moral dilemma multiple times (e.g., 10 times or more) to each model variant. This technique could improve the robustness of the findings by assessing the consistency and reliability of LLM responses, which provides deeper insights into how models handle repeated exposure to the same moral dilemma.

Finally, the relationship between abstract moral reasoning patterns and practical ethical applications is an important area for future research. Researchers may want to investigate whether LLMs that exhibit sophisticated moral reasoning in standardized assessments can effectively apply these skills to real-world ethical challenges (e.g., through AI agents and ChatGPT Atlas). In addition, understanding how moral reasoning moves from abstract scenarios to domain-specific applications is critical for creating AI systems capable of making ethical decisions in a variety of professional and personal settings.

13 Conclusion

This study shows that LLMs have sophisticated moral reasoning abilities that differ significantly from human patterns. The findings showed that LLMs consistently demonstrate increased post-conventional reasoning while relying less

on personal interests and conventional norm-based considerations. These systems are particularly responsive to prompt engineering and contextual interventions, demonstrating remarkable adaptability in their moral reasoning frameworks.

The findings also improve our understanding of AI's moral reasoning abilities while emphasizing key differences between AI and human ethical cognition. The higher moral reasoning scores observed in LLMs, while impressive, should be interpreted in light of their training processes, input data, and model architectures, which lead to major differences in their moral reasoning. In addition, the intervention results are especially significant because they show that LLMs can systematically adapt their moral reasoning when presented with historical moral exemplars. As a result, this adaptability, combined with the equivalence of various intervention approaches, suggests promising applications for moral education, while emphasizing the importance of carefully considering ethical implications.

These findings provide the groundwork for future research into AI moral reasoning and suggest strategies for developing more ethically aligned AI systems. However, incorporating such systems into moral education contexts needs careful consideration of their limitations and appropriate safeguards to ensure responsible and beneficial applications.

Author contributions Davin Nabizadeh (DN) conceptualized the study design, conducted all data collection across 32 LLM models and 9 platforms, performed statistical analyses using SPSS, interpreted findings, and wrote the main manuscript text including all sections from introduction through conclusion. David Walker (DW) provided supervisory guidance on research design, contributed to theoretical framework development (Four-Component Model and neo-Kohlbergian theory), reviewed the content for accuracy, and provided critical feedback on manuscript interpretation and implications. Hyemin Han (HH) contributed to the literature review, provided expertise on LLM moral reasoning capabilities and prompt engineering methodologies, reviewed intervention design based on moral exemplar research, and provided editorial feedback on manuscript revisions. Laura McNeill (LM) contributed to study design and structure, designed and refined prompt engineering strategies, reviewed and edited the manuscript, and provided critical refinement of the overall work and theoretical positioning. Stefanie Wind (SW) provided statistical consultation on comparative analysis approaches, reviewed effect size calculations and ANOVA/Mann-Whitney U test implementations, and offered guidance on data interpretation methodology. Jason Scofield (JS) contributed to the ethical implications discussion, reviewed responsible AI development recommendations, and provided feedback on educational and policy implications of findings. All authors reviewed the complete the study, agreed with the conclusions and implications, and approved the final version for submission.

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Declarations

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